

Head Band Camera

Technical Specification

Features

- **Imaging:** Full HD (1080p) streaming at 30 FPS H.264 over a standard, UVC USB interface.
- **Cross-Platform:**
Multi-mode transport layer natively supporting dual host topologies:
 - Android Architecture: Standard USB UVC interface paired with a mandatory, dedicated Android SDK.
 - iOS Architecture: High-efficiency USB CDC-NCM (Network Control Model) Ethernet emulation for seamless video/data streaming over native iOS networking layers.
- **Integrated IMU:**
Features an onboard IMU sensor tracking real-time linear acceleration, angular velocity at a high-frequency sampling rate.
- **Hardware-Level Sensor Fusion:**
Completely eliminates software-induced clock drift by triggering spatial timestamps precisely at the mid-exposure point of every single video frame.
Inertial metrics are output simultaneously with the video payload via structured data wrappers.
- **ISP Tuning:** Consumer-grade visual appeal, targeting color accuracy
Mean Delta E <2.5, sub-1.5 second Auto White Balance recovery.
- **Onboard Calibration Memory:** A dedicated non-volatile memory footprint accessible via custom UVC Extension Units (XU) or NCM control commands allows writing, reading, and locking geometric lens intrinsics and camera-to-IMU extrinsics.
- **Formfactor:** Ergonomic, Lightweight, head-strap-ready hardware form factor weighing under 35 grams (excluding head strap), bundled with a highly shielded, production-grade USB Type-C cable for unrestricted wearability.

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1. Imaging & Optical Specifications

Parameter	Specification	Technical Notes / Dynamic Adjustability
Camera Resolution	3 MP	Optimized pixel array size for high-velocity tracking.
Sensor Size/ Optical Format	1/2.8"	High-sensitivity sensor dimensions.
Shutter Type	Rolling Shutter Or Global shutter	Flexible Shall be optimized for eliminating motion blur and geometric distortion during rapid head movement.
Lens Optimization	Fixed Distance Lens focused at 1.5 meters.	Factory calibrated and locked to a static focal plane.
Horizontal Field of View (HFOV)	150°	Ultra-wide fisheye lens optimization.
Aspect Ratio	16:9	Cropped/scaled native output format.
Auto Focus	No	Lens is not autofocus/VCM
OIS	No	No optical image stabilization

2. Video Encoding & Streaming Specifications

Parameter	Specification	Technical Notes / Dynamic Adjustability
Primary Streaming Format	FHD (1920x1080) @ 30 FPS	Fixed native delivery frame rate.
Pixel Format	YUV 4:2:0	Native chroma subsampling format used by the H.264 hardware encoder for efficient compression
Video Compression	H.264 Encoding	Native hardware encoder engine.
Timestamping	Hardware	generated frame timestamps
Base Color Space	BT.709 / Rec.709	Standard color distribution configuration.
High Dynamic Range (HDR)	HDR on/off controllable SDR on	Optimized for consistent low-latency VIO processing.
Controllable Bitrate Delivery	7–10 Mbps (CBR)	Configurable constant bitrate from 7 Mbps to 10 Mbps.
GOP Length	30 Frames	Group of Pictures configurable via UVC Extension Units / NCM parameters.

3. Sensor Fusion & Synchronization (IMU)

Parameter	Specification	Operational Behavior
IMU DOF	6DOF minimum	Accelerometer, Gyroscope.
Jitter	10us	Jitter shall be under 10us
Sampling Frequency	250Hz - 400 Hz	High-rate inertial state tracking.
Hardware Synchronization	Precision Frame-Synced	Middle of image exposure for global shutter, Middle of first row exposure time for rolling shutter camera
Data Architecture	SDK-Driven Payload Access	Interleaved inertial state frames are read programmatically over the UVC/NCM stream using the platform SDK.
IMU - Camera Extrinsic	sub-degree, sub-mm accuracy	Shall be sub mm accuracy on the values from hardware.

4. Connectivity & System Integration

Parameter	Specification	Implementation Requirements
USB Class Protocols	Dual-Mode Firmware (UVC & CDC-NCM)	Android/PC Mode: Standard USB UVC video device profile. iOS Mode: USB Communications Device Class Network Control Model (CDC-NCM) for Ethernet emulation.
Physical Interface	USB Type-C	Reversible industrial connector integration.
Android Integration	Mandatory Native Android SDK	The system must deliver full feature parity (frame acquisition, camera parameters, calibration access) via a custom Android SDK wrapper utilizing standard Android UVC API hooks.
iOS Integration	Native iOS NCM Ethernet Camera Mode	Camera must expose a compliant network device interface to iOS hosts, bypassing standard UVC mobile restrictions via IP networking protocols.
Cabling Specifications	Production-Grade USB Type-C	Length: 1.5 m USB 3.2 Gen 1 (5 Gbps) compliant Backward compatible with USB 2.0 Controlled 90 Ω differential impedance for SuperSpeed pairs Shielded construction for EMI reduction and signal integrity Minimum 24 AWG conductors for VBUS and GND to minimize voltage drop
Physical Design / Weight	Lightweight Head-Band Integrated	Under 35 grams (Structural weight metric excludes the removable head strap).

5. Camera Image Controls

These commands are exposed natively via the standard Video Processing Unit (PU) and Camera Terminal (CT) descriptors for UVC hosts, and mapped to matching configuration packets for NCM hosts.

- **Auto Exposure Mode (AE) (R/W):**
Must support dynamic switching between Auto Mode and Manual Mode.
- **Exposure Time (Absolute) (R/W):** r.
Adjustable exposure time with range that prevents motion blur under bright environments.
- **Brightness (R/W):**
Modifies digital black level offsets. Range: Standard -64 to +64 or 0 to 255.
- **Contrast (R/W):**
Manages Gamma transfer curve slope definitions. Range: 0 to 255.
- **Gain (R/W):**
Hardware Analog/Digital multiplier. Must be lockable when Exposure Mode is Manual.
- **Saturation (R/W):**
Image chroma intensity scale. Setting to 0 must result in true grayscale output. Max range up to 200%.
- **Sharpness (R/W):**
Manages the internal high-pass edge enhancement filter within the ISP.
- **White Balance Temperature (R/W):**
Supports Auto (AWB) or Manual Kelvin selection (Range: 2800K to 6500K).
- **White Balance Component (R/W):**
Direct software balance control over individual Red and Blue raw digital channels.
- **Power Line Frequency (R/W):**
Hardware anti-flicker selection. Selectable options: 50 Hz (Europe/Asia) or 60 Hz (Americas).

6.UVC Extension Unit [XU] / CDC Network Controls

For proprietary configuration control over custom sensor, compression, and calibration features across both Android (XU) and iOS (Network Commands) SDK variants.

- **Encoder Bitrate Control (R/W | 4 Bytes):**
- **GOP Length (R/W | 2 Bytes):**
- **IMU Stream Control (R/W | 1 Byte):**
- **Calibration Protect (R/W | 1 Byte):**
- **Lens Intrinsic Data (R/W | 64 Bytes):**
- **System Extrinsic Data (R/W | 64 Bytes):**

6.1 Calibration Intrinsic/Extrinsic

The camera needs to be calibrated for intrinsic (including lens distortion) parameters, and for Rolling Shutter cameras the readout time parameters. Stored as json structures on board

7. Image Quality (IQ)

The factory ISP tuning must validate against these precise target performance standards:

IQ Parameter	Test Method / Chart	Test Conditions	Target Specification
Color Accuracy	24-Patch ColorChecker ISO 17321	D65 (Daylight), 1000 Lux	Mean Delta E $0 < 2.5$, Max Delta E $0 < 5.5$
Color Saturation	24-Patch ColorChecker	Across all standard light sources	105% +/- 10% Chroma level
Skin Tone Preservation	24-Patch Chart / Live Subjects	Multi-ethnic skin tone profiles	Delta E $0 < 2.0$ (Strict skin tone deviation limits)
White Balance Accuracy	6-Step Gray Scale Chart	D65, TL84, Horizon, Tungsten A	Delta C < 1.5 (Zero visible color cast on neutral patches)
White Balance Error	6-Step Gray Scale Chart	Uniform continuous light cells	Delta C < 1.0
AWB Transition Speed	Lux/Color changing light box	Instant shift between 2700K & 6500K	Fully settled and stabilized within 1.5 seconds
Auto Exposure	The Light Booth	Dynamic step-change in Lux	$< 150\text{ms}$

Color Uniformity	Uniform Flat-Field Target	Maximum Lens Aperture	Max corner-to-center variance Delta E 0 < 2.0
Color Shading Uniformity	Uniform Flat-Field Target	Maximum Lens Aperture	Max corner-to-center deviation Delta E 0 < 1.5

8. Thermal & Environmental

Temperature Ranges

The system must operate safely and maintain high image quality within standard consumer electronics environments, with an expanded thermal safety window for storage.

Parameter	Specification	Technical Conditions / Notes
Operating Temperature	0°C to +50°C	Ambient air temperature. Fully operational within all stated IQ specs.
Storage Temperature	-20°C to +70°C	Non-operating state. Safe for transport and warehouse storage environments.
Max Skin Contact Temp	< 43C	Critical Wearable Safety Constraint. The outer chassis touching the user's skin must never exceed this threshold during continuous operation at 50°C ambient to prevent tactile burns.

9. Firmware Update

Device firmware shall be updateable over the USB Type-C interface using the provided Software Development Kit (SDK).

10.Audio Microphone (USB Audio Class)

1. Audio Capture & Encoding

Parameter	Specification	Technical Notes / Implementation
Interface Protocol	USB UAC 1.0/2.0	Configured as part of the USB Composite Device descriptor.
Microphone Type	Digital MEMS Microphone	High SNR omnidirectional microphone array.
Audio Format	PCM Linear (Uncompressed)	Standard, low-latency raw audio stream.
Sampling Rate	48 kHz (Default) / 16 kHz	Software selectable via standard OS audio controls.
Bit Depth	16-bit or 24-bit Mono	Optimized for voice clarity and low bandwidth consumption.
Synchronization	Asynchronous Endpoint	The camera's hardware internal clock drives the audio sample delivery rate to prevent audio-video drift.
Mute Control	On/Off hardware	Controllable over SDK, when mute hardware enabled.